CYBERSECURITY COURSE · SIEM PROJECT

Splunk Enterprise Host Monitoring

Part 1: install, monitor and investigate a Windows host

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Machine-specific identifiers (SIDs) redacted in screenshots

SCOPE

Objective

- **1.** Install Splunk Enterprise on a host
- **2.** Monitor the host's OS logs and performance data
- **3.** Build dashboards from the monitored data
- **4.** Write up the setup, technology, findings and lessons learned

Approach

- Prepare Windows auditing so the right events are logged
- Ingest Security, System and Application logs plus CPU and memory counters
- Generate test activity (account creation, failed logins, lockout)
- Build a two-tab dashboard (Performance | Security) and investigate anything unusual

SECTION 1

Setup & Technology

A SIEM aggregates events, performance and audit data from different sources into a single pane of glass, so analysts can track and investigate activity efficiently.

Host

- Acer Nitro AN515-43
- AMD Ryzen 7 3750H · 32 GB RAM · 477 GB SSD
- Windows 11 Home 25H2 (build 26200.9711), 64-bit

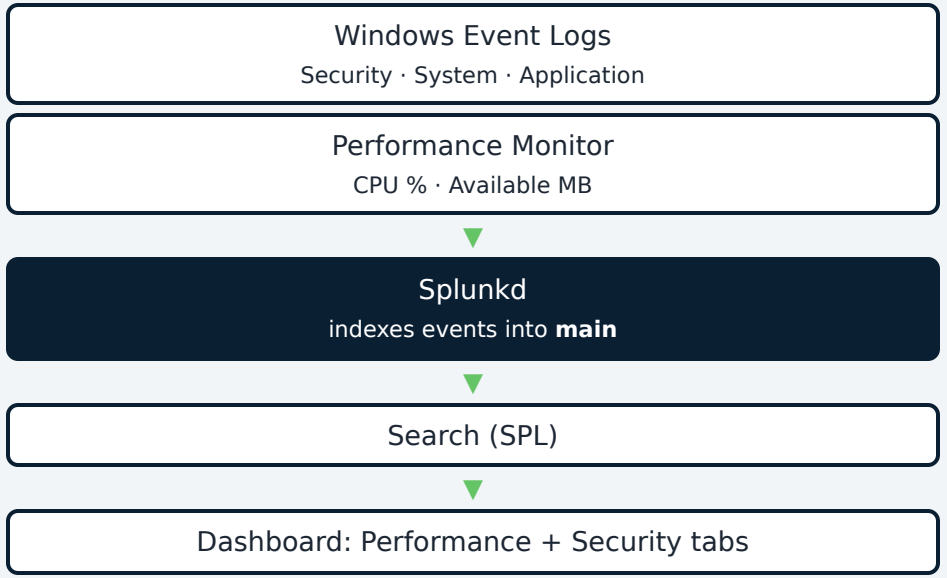
SIEM

- Splunk Enterprise 10.6.0.5, standalone instance
- Trial licence → free licence, 500 MB/day
- Search language: SPL · dashboards: Dashboard Studio

Windows tools used

- GUI: services.msc, Event Viewer
- CMD: auditpol, net accounts, net user, powercfg

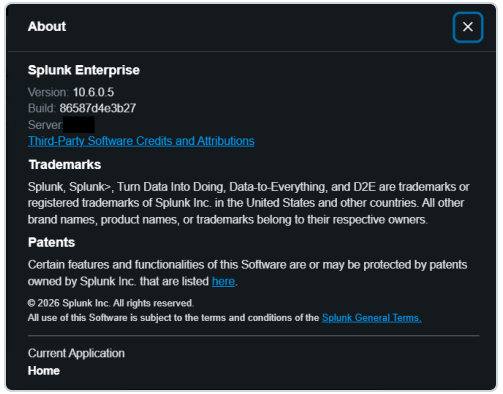
Data flow



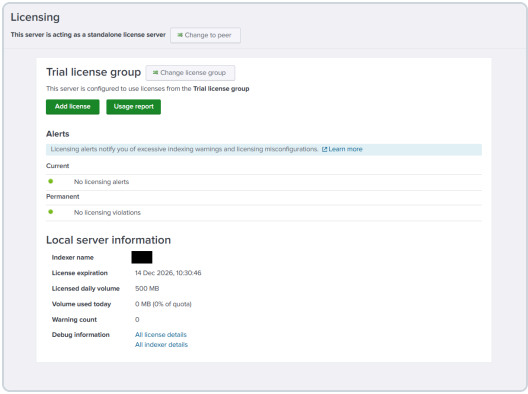
Resources: Splunk Docs, Splunk Community, web research, and an AI assistant (Claude) to explain concepts, speed up SPL syntax and help structure this write-up. All installation, configuration, testing and investigation were done by me.

Installation

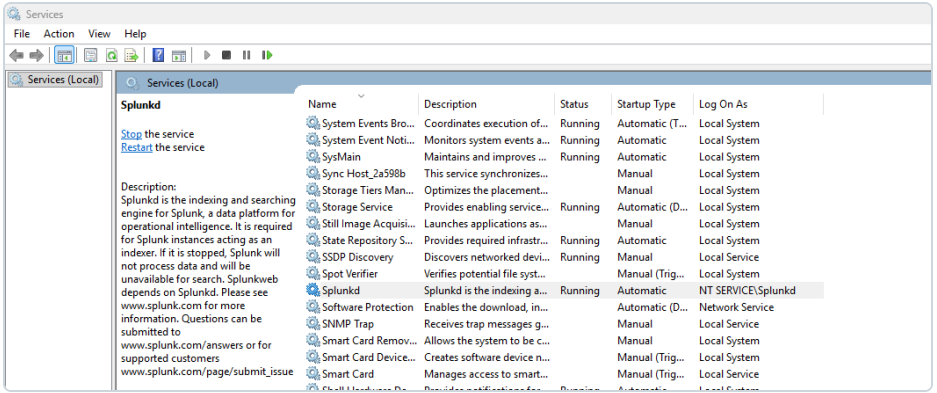
- Installed Splunk Enterprise 10.6.0.5 via .msi on host host
- Web UI at **localhost:8000**
- Service runs as **NT SERVICE\Splunkd**, a least-privilege virtual account, not Local System
- Trial licence active: 500 MB/day, no violations



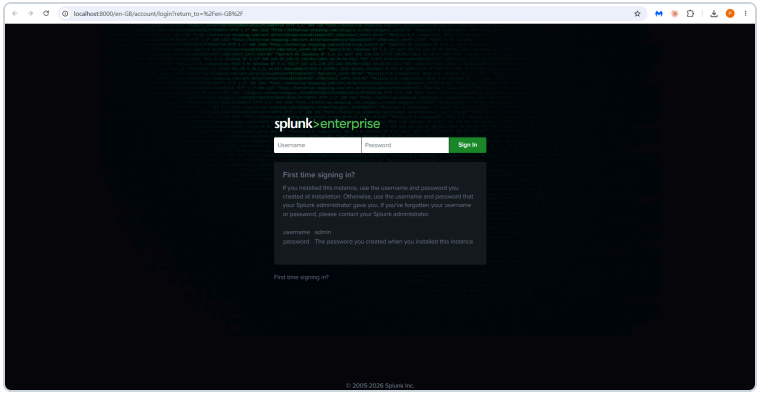
Version and build



Licence: 0 MB used before ingestion



Splunkd running as NT SERVICE\Splunkd



Web UI login

SECTION 1

Host pre-configuration

- Checked the audit policy with `auditpol /get /category:*`
- Logon: Success & Failure ✔ · User Account Management: Success ✔ · Security Group Management: Success ✔
- Lockout policy (`net accounts`): 10 attempts / 10 min; lowered to **3** for the lockout test
- Windows **Home** has no gpedit.msc, so all policy work was done from the command line

```
C:\Windows\System32>net accounts
Force user logoff how long after time expires?:       Never
Minimum password age (days):                          0
Maximum password age (days):                          42
Minimum password length:                              0
Length of password history maintained:                None
Lockout threshold:                                    10
Lockout duration (minutes):                           10
Lockout observation window (minutes):                 10
Computer role:                                        WORKSTATION
The command completed successfully.


C:\Windows\System32>
```

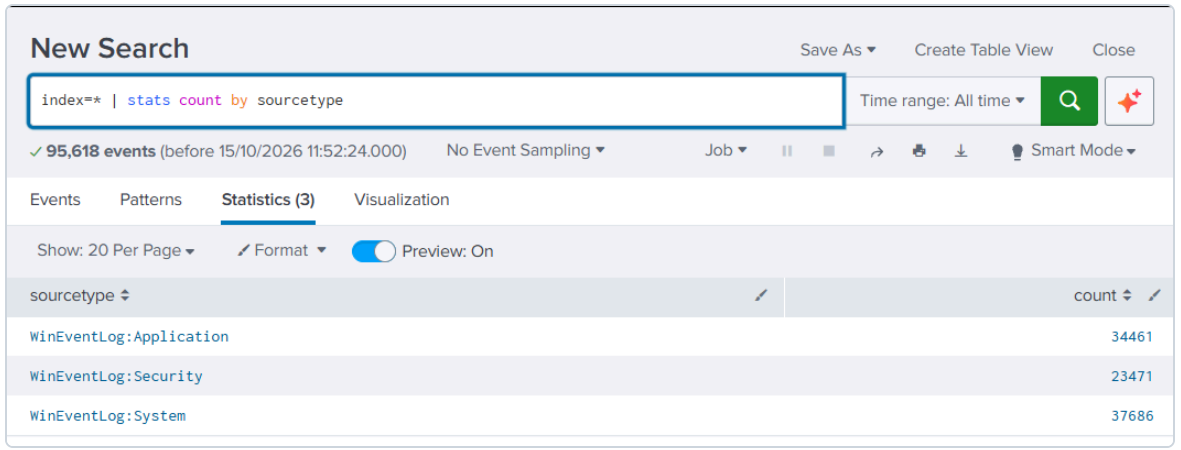
Machine-wide password and lockout policy

```
C:\Windows\System32>auditpol /get /category:*
System audit policy
Category/Subcategory                      Setting
System
  Security System Extension               No Auditing
  System Integrity                        Success and Failure
  IPsec Driver                            No Auditing
  Other System Events                     Success and Failure
  Security State Change                   Success
Logon/Logoff
  Logon                                   Success and Failure
  Logoff                                  Success
  Account Lockout                         Success
  IPsec Main Mode                         No Auditing
  IPsec Quick Mode                        No Auditing
  IPsec Extended Mode                     No Auditing
  Special Logon                           Success
  Other Logon/Logoff Events               No Auditing
  Network Policy Server                   Success and Failure
  User / Device Claims                    No Auditing
  Group Membership                        No Auditing
  Access Rights                           No Auditing
Object Access
  File System                             No Auditing
  Registry                                No Auditing
  Kernel Object                           No Auditing
  SAM                                     No Auditing
  Certification Services                  No Auditing
  Application Generated                   No Auditing
  Handle Manipulation                     No Auditing
  File Share                              No Auditing
  Filtering Platform Packet Drop          No Auditing
  Filtering Platform Connection           No Auditing
  Other Object Access Events              No Auditing
  Detailed File Share                     No Auditing
  Removable Storage                       No Auditing
  Central Policy Staging                  No Auditing
Privilege Use
  Non Sensitive Privilege Use             No Auditing
  Other Privilege Use Events              No Auditing
  Sensitive Privilege Use                 No Auditing
Detailed Tracking
  Process Creation                        No Auditing
  Process Termination                     No Auditing
  DPAPI Activity                          No Auditing
  RPC Events                              No Auditing
  Plug and Play Events                    No Auditing
  Token Right Adjusted Events             No Auditing
Policy Change
  Audit Policy Change                     Success
  Authentication Policy Change            Success
  Authorization Policy Change             No Auditing
  MPSSVC Rule-Level Policy Change         No Auditing
  Filtering Platform Policy Change        No Auditing
  Other Policy Change Events              No Auditing
Account Management
  Computer Account Management             No Auditing
  Security Group Management               Success
  Distribution Group Management           No Auditing
  Application Group Management            No Auditing
  Other Account Management Events         No Auditing
```

Audit policy (before changes)

Data inputs

- Inputs configured manually via Settings → Data inputs (event logs + performance counters)
- **Event logs:** Security, System, Application → indexed into main
- **Performance:** Processor % (_Total) and Memory Available MBytes, polled every 20 s
- Verified ingestion with `index=* | stats count by sourcetype`
- Security log readable by the low-privilege service account: the installer granted it Event Log Readers (see Finding 3)



About 95k events ingested across the three logs

Event Log collection name	Log(s)	Host(s)	Index	Status	Actions
localhost	Application, Security, System	localhost	default	Enabled | Disable	

Event log inputs: Application, Security, System

Collection name	Counters	Instances	Polling interval	Index	Status	Actions
CPU	% Processor Time	_Total	20	default	Enabled | Disable	Clone | Delete
Memory	Available MBytes	*	20	default	Enabled | Disable	Clone | Delete

Performance inputs: CPU % and Available Memory, polled every 20 s

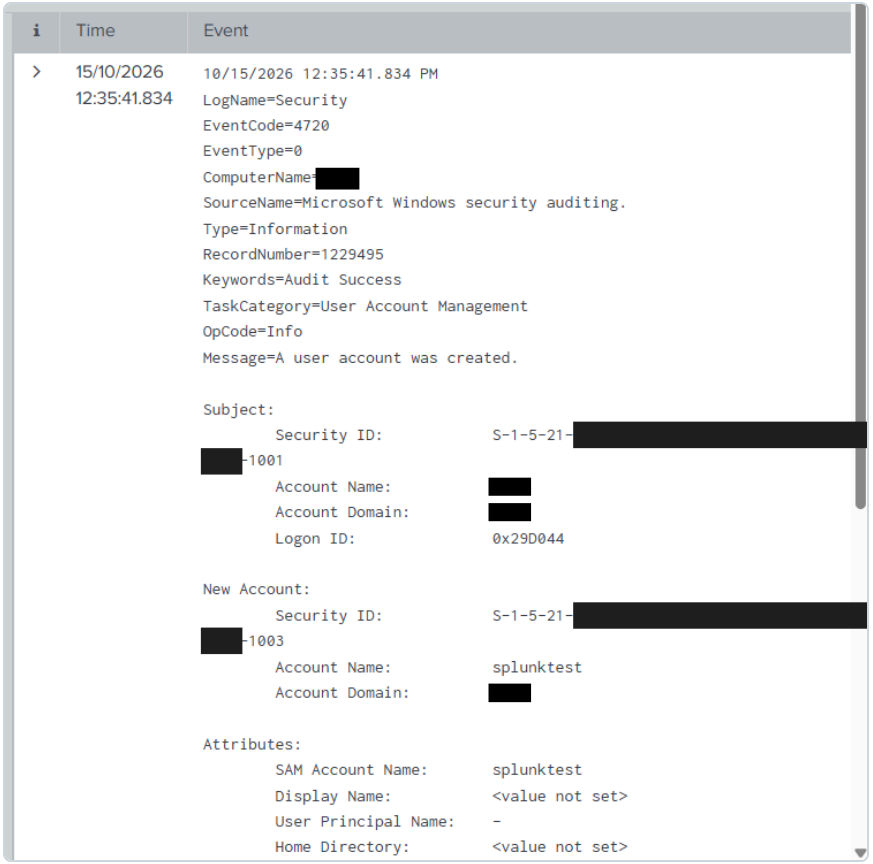
SECTION 1

Simulated activity

Action	Command	Events
Create test user	`net user splunktest <pw> /add`	4720 + 4732 (added to Users as part of the command)
Failed logins (sign-in screen)	wrong password ×6	4625 ×6 (13:58)
Lockout test (threshold 3)	`runas /user:splunktest cmd` ×3	4625 ×3 → **4740** (18:35:45)

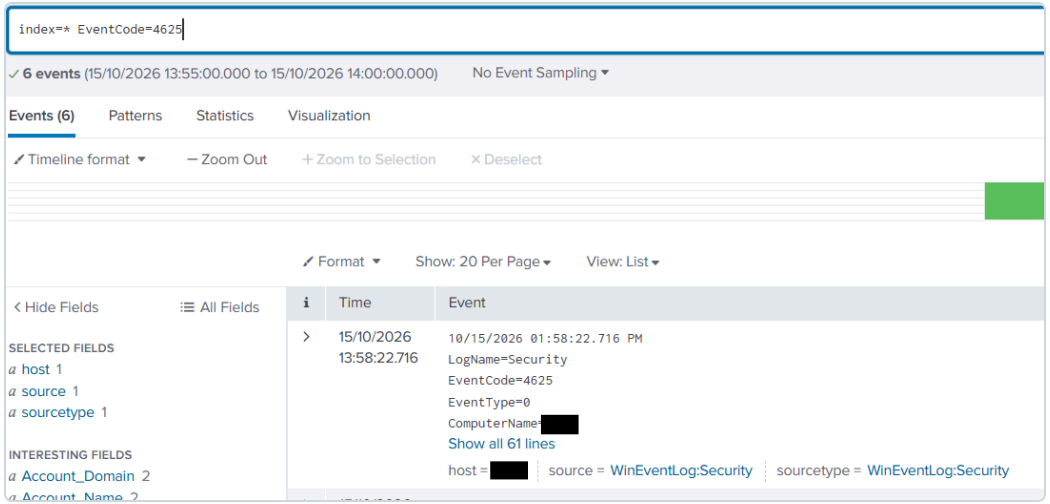
The 6 sign-in attempts triggered the Windows sign-in screen's own throttling before the lockout threshold of 10 was reached, so no lockout occurred. I lowered the threshold to 3 and retested via runas, which produced the 4740.

Each event was checked in Splunk straight after the action. Expanding an event shows **Subject** (who did it) and the target (who it was done to).

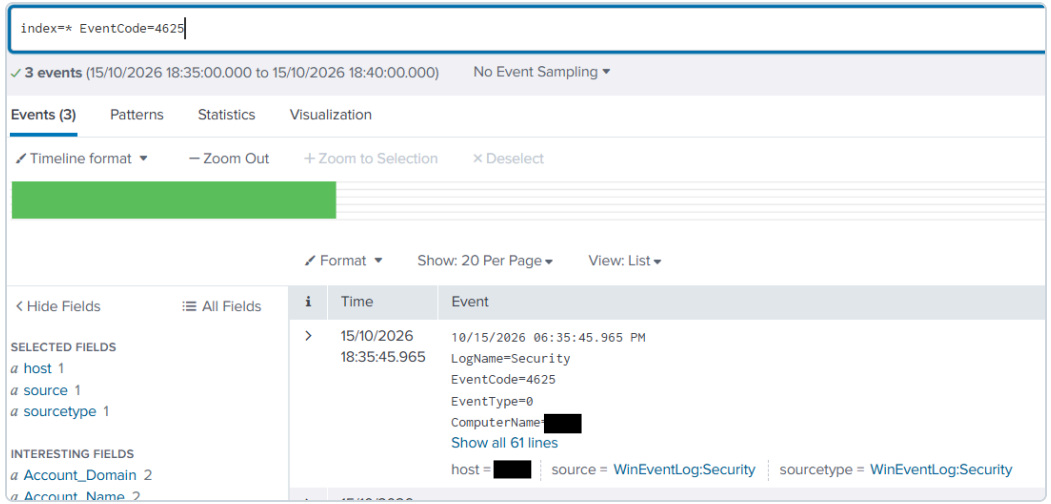


4720: host created splunktest

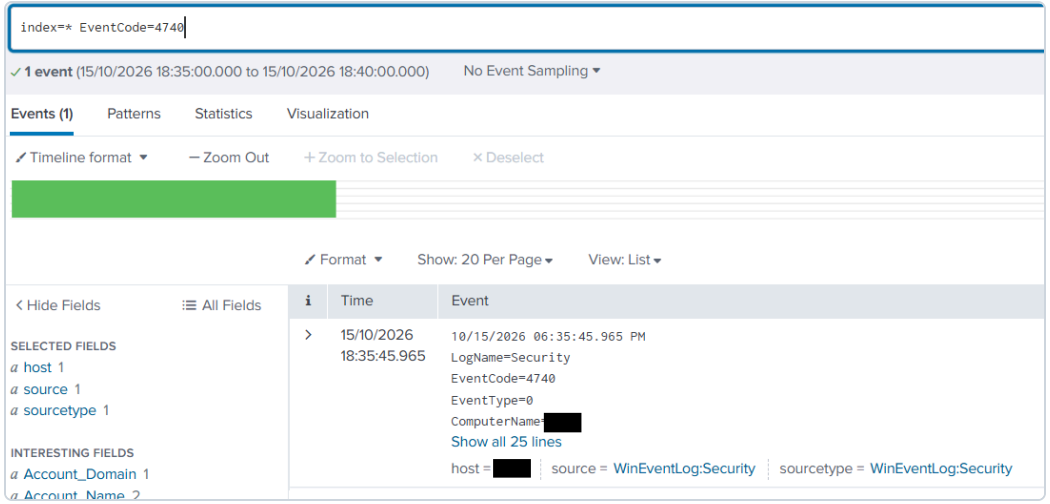
Simulated activity · evidence



6 × 4625 at 13:58: sign-in screen test



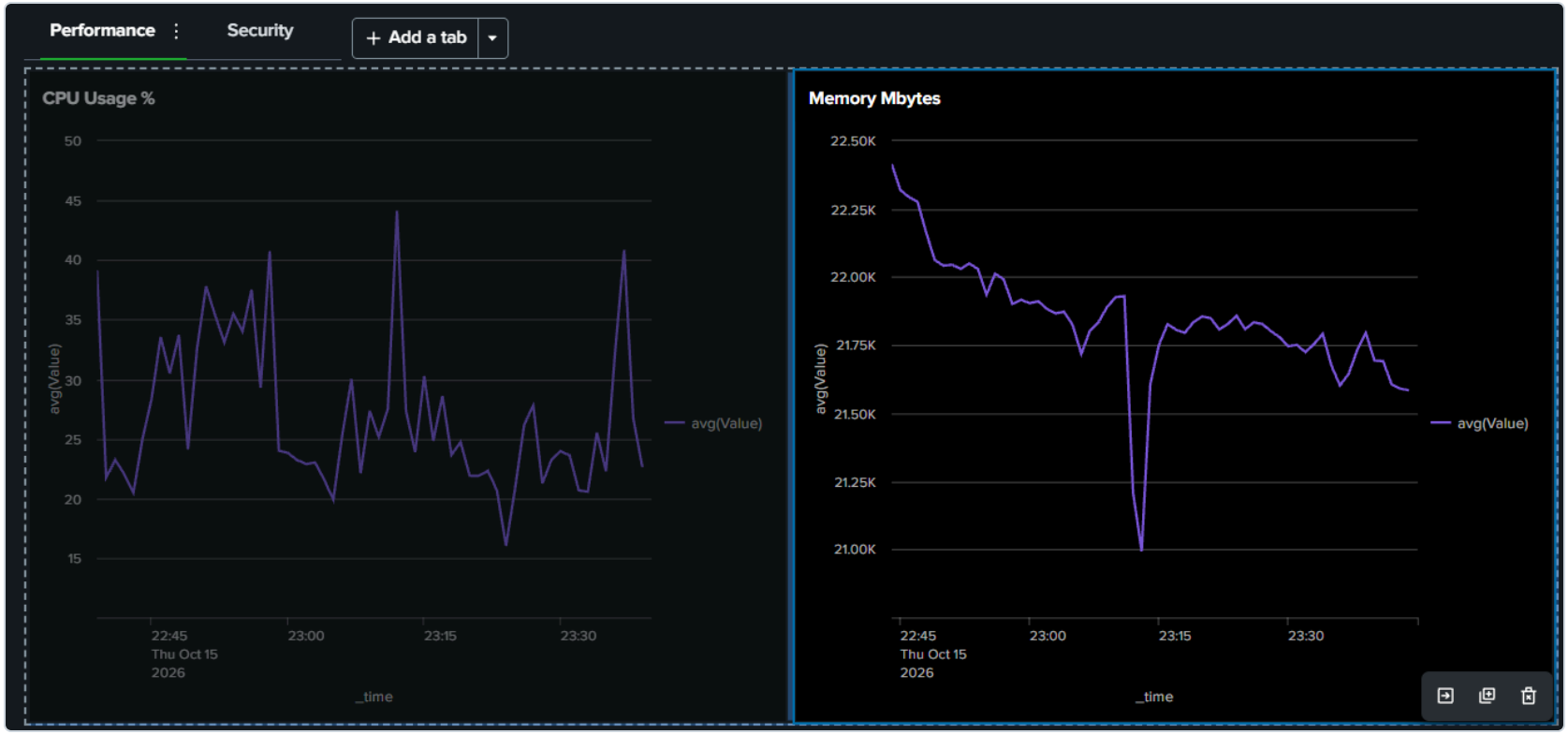
3 × 4625 at 18:35: runas test, threshold 3



4740 at 18:35:45.965: account locked out

The third failed login and the lockout share the exact same timestamp, **18:35:45.965**: the third wrong password triggered the lock instantly, proving the threshold of 3 worked as configured.

Dashboard · Performance tab



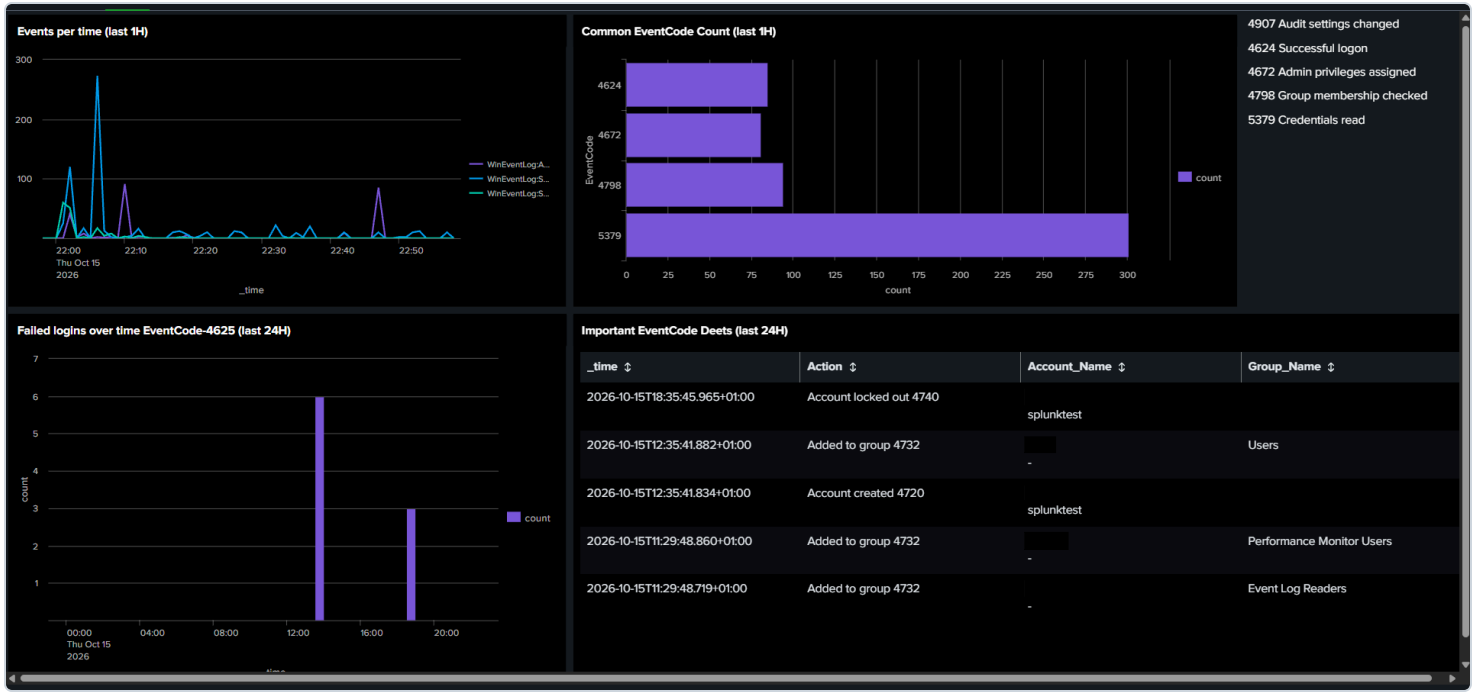
CPU Usage % and Available Memory (MB): 1-minute averages, auto-refresh enabled

```
source="Perfmon:CPU" | timechart span=1m avg(Value)
```

```
source="Perfmon:Memory" | timechart span=1m avg(Value)
```

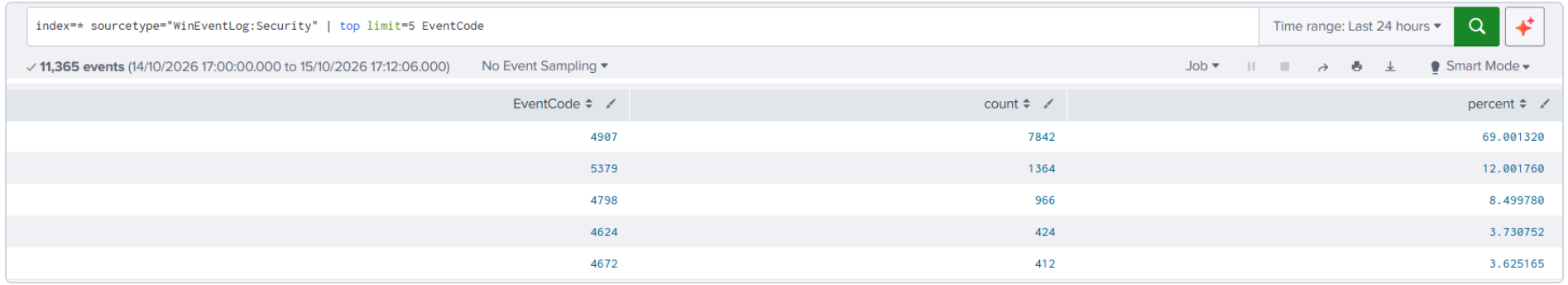
Baseline: CPU 15–45%, about 21–22.8 GB RAM free. **span=1m** averages the 3 readings per minute (20 s polling) and removes gaps.

Dashboard · Security tab



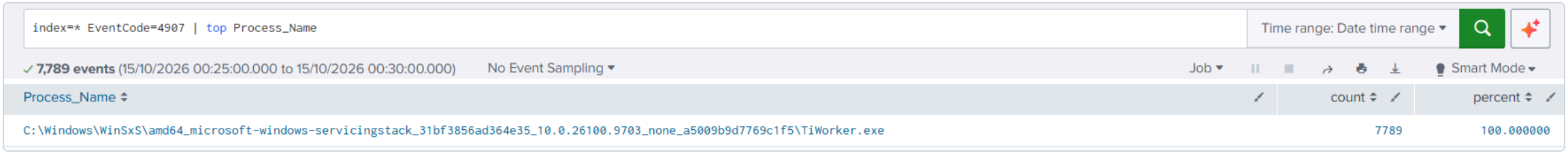
Panel	SPL
Events per time	`index=* sourcetype=WinEventLog* | timechart count by sourcetype`
Common event codes	`index=* EventCode IN (4907,5379,4798,4624,4672) | stats count by EventCode`
Failed logins	`index=* sourcetype="WinEventLog:Security" EventCode=4625 | timechart count`
Key account events	`index=* sourcetype="WinEventLog:Security" EventCode IN (4720,4732,4740) | eval Action=case(EventCode=4720,"Account created 4720", EventCode=4732,"Added to group 4732", EventCode=4740,"Account locked out 4740") | table _time Action Account_Name Group_Name`

Finding 1 · One event type was 69% of the Security log



EventCode	count	percent
4907	7842	69.001320
5379	1364	12.001760
4798	966	8.499780
4624	424	3.730752
4672	412	3.625165

Top 5 Security event codes, last 24 h

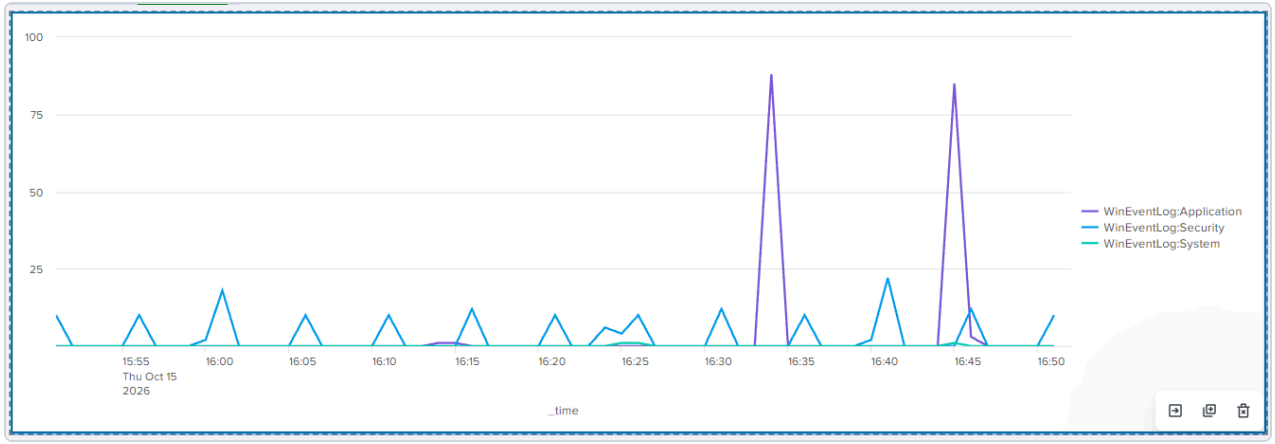


Process_Name	count	percent
C:\Windows\WinSxS\amd64_microsoft-windows-servicingstack_31bf3856ad364e35_10.0.26100.9703_none_a5009b9d7769c1f5\TiWorker.exe	7789	100.000000

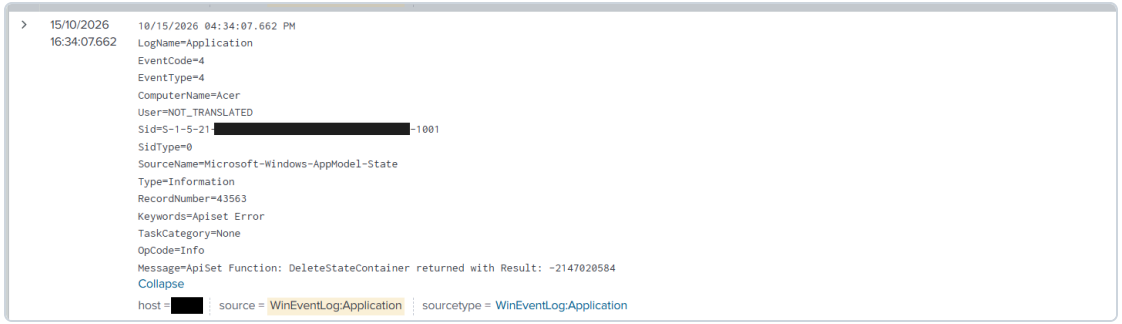
4907 burst: 7,789 events, 100% TiWorker.exe

- **4907** (auditing settings on an object changed): 7,842 of 11,365 Security events in 24 h
- The burst: **7,789 events in 5 minutes** (00:25–00:30), **100% from TiWorker.exe**
- TiWorker.exe = Windows Modules Installer Worker, the process that installs Windows updates
- Running as **SYSTEM** (S-1-5-18) from its legitimate WinSxS servicing-stack path, not a lookalike
- Files changed were .NET Framework components: one event per file as the update reapplied their security settings
- **Conclusion:** overnight Windows Update servicing, benign

Finding 2 · Application spikes and background noise



Events per time: Application spikes at 16:33 and 16:45



Drill-down: Microsoft-Windows-AppModel-State, Event 4

Application spikes (16:33, 16:45)

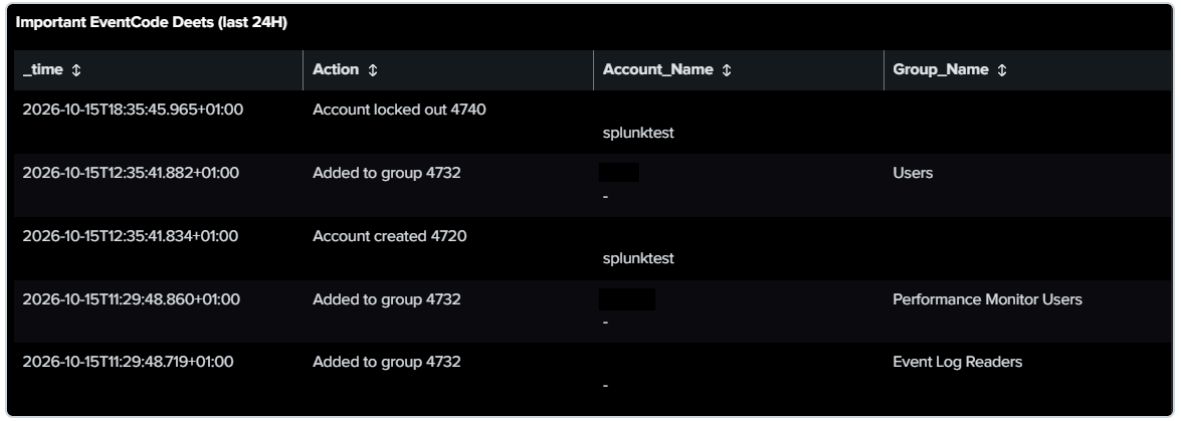
- About 85 events per spike from **AppModel-State** (packaged-app state cleanup)
- Error −2147020584 = 0x800710D8, **object not found**: deleting something already gone
- Triggered in the user session (SID …-1001), likely an app closing or updating, **benign**

Regular Security background

- **4798** (group membership checked) at about 120/h is the most common Security event in the 1-hour view

Finding 3 · Account activity

- Lockout control works: 3 failures → **4740** at 18:35:45, logged by HOST $ (the system)
- Creating splunktest (SID …-1003) also added it to **Users**, both logged under host 's session (4720 + 4732 at 12:35:41)
- **Unplanned 4732s at 11:29:48** by SYSTEM: Splunk's installer added its service account (Member SID **S-1-5-80-…-1203010618** = NT SERVICE\Splunkd) to **Event Log Readers** and **Performance Monitor Users**
- These are the minimum rights needed to read the protected Security log and performance counters, instead of running with admin rights. That's least privilege in practice. Without Event Log Readers, System and Application events would still be collected, but Security events would not.



Important EventCode Deets (last 24H)

_time ↕	Action ↕	Account_Name ↕	Group_Name ↕
2026-10-15T18:35:45.965+01:00	Account locked out 4740	splunktest	
2026-10-15T12:35:41.882+01:00	Added to group 4732	[redacted] -	Users
2026-10-15T12:35:41.834+01:00	Account created 4720	splunktest	
2026-10-15T11:29:48.860+01:00	Added to group 4732	[redacted] -	Performance Monitor Users
2026-10-15T11:29:48.719+01:00	Added to group 4732	-	Event Log Readers

Key account events panel (last 24 h)

Finding 3 · Evidence: the 4732 events

15/10/2026 11:29:48.860

```
10/15/2026 11:29:48.860 AM
LogName=Security
EventCode=4732
EventType=0
ComputerName=[redacted]
SourceName=Microsoft Windows security auditing.
Type=Information
RecordNumber=1228860
Keywords=Audit Success
TaskCategory=User Account Management
OpCode=Info
Message=A member was added to a security-enabled local group.

Subject:
        Security ID:            S-1-5-18
        Account Name:           [redacted]
        Account Domain:         WORKGROUP
        Logon ID:               0x3E7

Member:
        Security ID:            S-1-5-80-3191021196-3668848460-1609882449-918409182-1203010618
        Account Name:           -

Group:
        Security ID:            S-1-5-32-558
        Group Name:             Performance Monitor Users
        Group Domain:           Builtin

Additional Information:
        Privileges:             -
```

Collapse

host = [redacted] | source = WinEventLog:Security | sourcetype = WinEventLog:Security

11:29:48: SYSTEM adds Splunkd's SID (S-1-5-80-…) to Performance Monitor Users

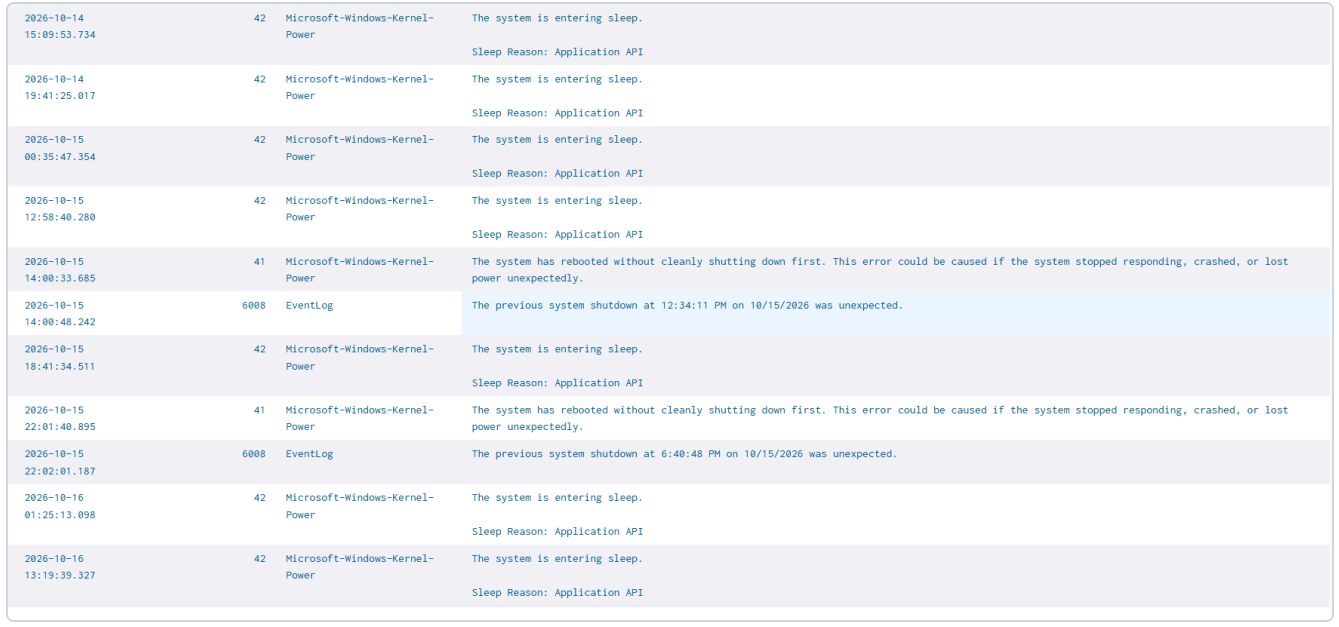
Time	Event
15/10/2026 12:35:41.882	10/15/2026 12:35:41.882 PM LogName=Security EventCode=4732 EventType=0 ComputerName=[redacted] SourceName=Microsoft Windows security auditing. Type=Information RecordNumber=1229499 Keywords=Audit Success TaskCategory=User Account Management OpCode=Info Message=A member was added to a security-enabled local group. Subject: Security ID: S-1-5-21-[redacted]-1001 Account Name: [redacted] Account Domain: [redacted] Logon ID: 0x29D044 Member: Security ID: S-1-5-21-[redacted]-1003 Account Name: - Group: Security ID: S-1-5-32-545 Group Name: Users Group Domain: Builtin Additional Information: Privileges: - Collapse host = [redacted] | source = WinEventLog:Security | sourcetype = WinEventLog:Security

12:35:41: host adds splunktest (…-1003) to Users

Finding 4 · Incident: system freezes around sleep

Time (15 Oct)	Event	Meaning
12:58	42	Sleep
14:00	41 + 6008	Forced off after a freeze (sign-in screen hung after resume)
18:41	42	Sleep
19:20–19:45	none (observed)	Fans at full speed, black screen, unresponsive
22:01	41 + 6008	Forced off and restarted

- **Event 41 details:** BugcheckCode 0 = a freeze, not a crash; SleepInProgress 4 = froze during a power transition, read as hibernate
- **Chronic, not a one-off:** 42 Event 41s since 24 Feb 2026, visible in the host's own logs and beyond Splunk's 2-day window
- **Side effect:** Splunk had a monitoring gap and repaired its data on the next start



2026-10-14 15:09:53.734	42	Microsoft-Windows-Kernel-Power	The system is entering sleep. Sleep Reason: Application API
2026-10-14 19:41:25.017	42	Microsoft-Windows-Kernel-Power	The system is entering sleep. Sleep Reason: Application API
2026-10-15 00:35:47.354	42	Microsoft-Windows-Kernel-Power	The system is entering sleep. Sleep Reason: Application API
2026-10-15 12:58:40.280	42	Microsoft-Windows-Kernel-Power	The system is entering sleep. Sleep Reason: Application API
2026-10-15 14:00:33.685	41	Microsoft-Windows-Kernel-Power	The system has rebooted without cleanly shutting down first. This error could be caused if the system stopped responding, crashed, or lost power unexpectedly.
2026-10-15 14:00:48.242	6008	EventLog	The previous system shutdown at 12:34:11 PM on 10/15/2026 was unexpected.
2026-10-15 18:41:34.511	42	Microsoft-Windows-Kernel-Power	The system is entering sleep. Sleep Reason: Application API
2026-10-15 22:01:40.895	41	Microsoft-Windows-Kernel-Power	The system has rebooted without cleanly shutting down first. This error could be caused if the system stopped responding, crashed, or lost power unexpectedly.
2026-10-15 22:02:01.187	6008	EventLog	The previous system shutdown at 6:40:48 PM on 10/15/2026 was unexpected.
2026-10-16 01:25:13.098	42	Microsoft-Windows-Kernel-Power	The system is entering sleep. Sleep Reason: Application API
2026-10-16 13:19:39.327	42	Microsoft-Windows-Kernel-Power	The system is entering sleep. Sleep Reason: Application API

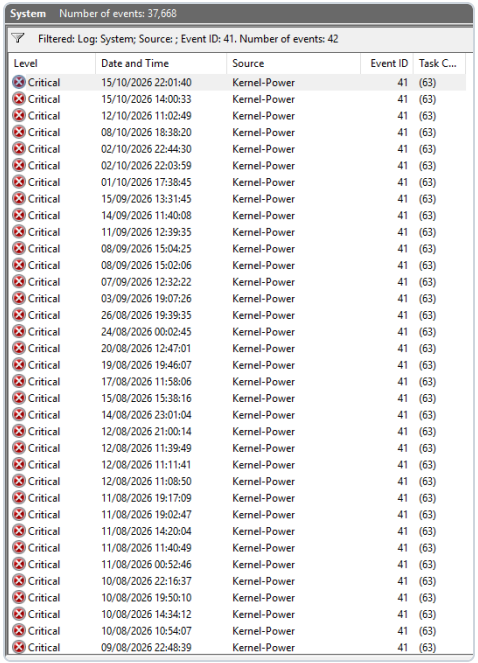
Splunk: System log 41 / 42 / 6008, sorted by time

```
Event 41 · 15/10/2026 22:01:40 (Event Viewer → Details)
BugcheckCode          0
SleepInProgress       4
PowerButtonTimestamp  0
ConnectedStandbyInProgress  false
```

Finding 4 · Investigation & remediation

Hypothesis	Test	Result
Blue-screen crash	Event 41 BugcheckCode	✗ 0 = freeze
Modern Standby	`powercfg /a`	✗ S3 sleep only
Scheduled wake	`powercfg /waketimers`	✗ none
Hibernate timer	`powercfg /q … HIBERNATEIDLE`	✗ never on AC, 3 h on battery
Critical battery	`powercfg /batteryreport`	✗ 90–100%
Outdated BIOS	BIOS version check	✗ V1.12, already the latest
Outdated graphics driver	Device Manager	⚠ AMD driver dated 14/07/2020 → updated 16 Oct

- **Suspected cause:** a driver or firmware failure during the hibernate power transition; the 2020 AMD graphics driver on Windows 11 25H2 is the leading suspect
- **Remediation:** AMD graphics driver updated to Adrenalin 26.5.2 (16 Oct); hibernate left enabled so only one variable changed; Process Creation auditing enabled to capture what runs before any recurrence
- **Observed:** desktop and apps noticeably more responsive after the update
- **Status:** monitoring for new Event 41s (excluding the forced restart after the install, 16 Oct 20:55); if freezes continue, the next single test is hibernate disabled



Event Viewer: 42 × Event 41 since Feb 2026

```
Device Manager · AMD Radeon RX Vega 10 Graphics
Driver date before:  14/07/2020
Driver date after:   20/05/2026  (Adrenalin 26.5.2)
```

Finding 5 · Host hardening gaps

Gaps

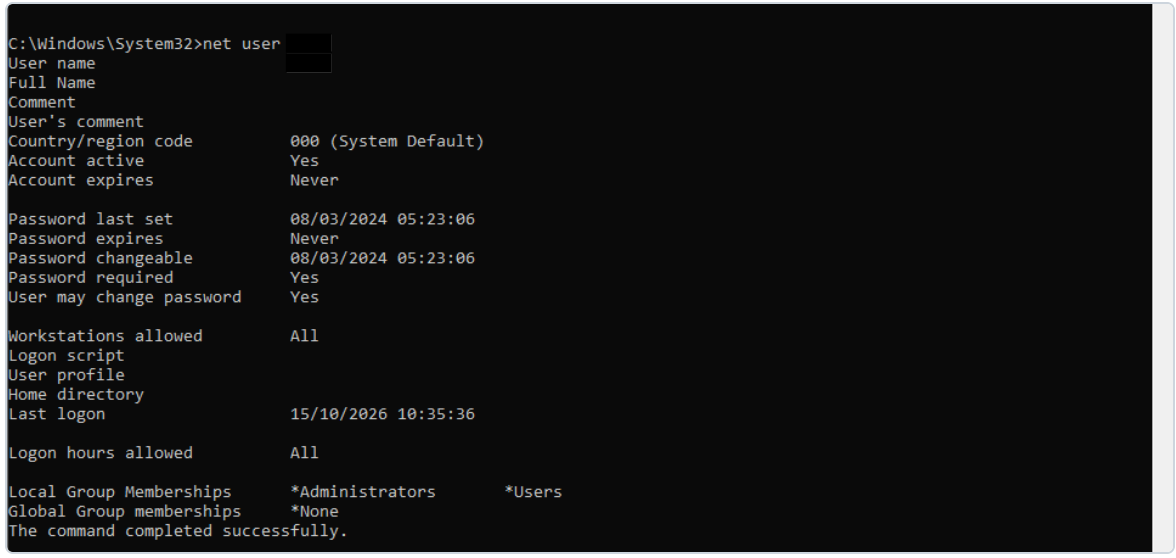
- **Daily account is an Administrator:** anything run in the session gets full rights (breaks least privilege)
- **Password never expires:** last set 08/03/2024, although the policy says 42 days. The account is exempt (policy vs reality)
- **Minimum password length 0, no password history**
- **Test account still enabled:** splunktest has a simple password and is active after testing
- **Process Creation auditing off by default:** no record of what ran on the machine (now enabled)

Good

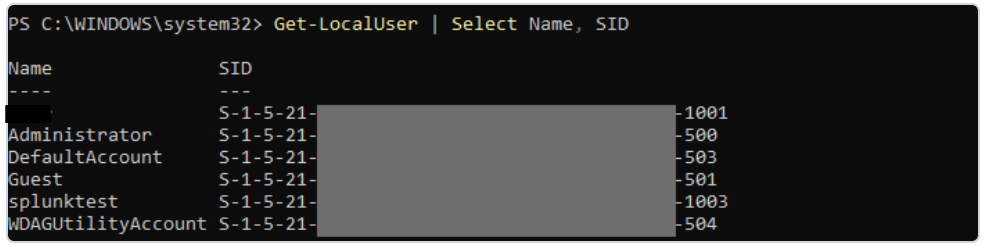
- Built-in Administrator, Guest, DefaultAccount and WDAGUtilityAccount all disabled ✔ (only host and splunktest enabled)

Observation

- RID 1002 is missing between host (1001) and splunktest (1003): an account was created after host and later deleted. Possibly from the previous owner (laptop bought second-hand) or a temporary Windows account. Gaps in the RID sequence reveal accounts even after deletion

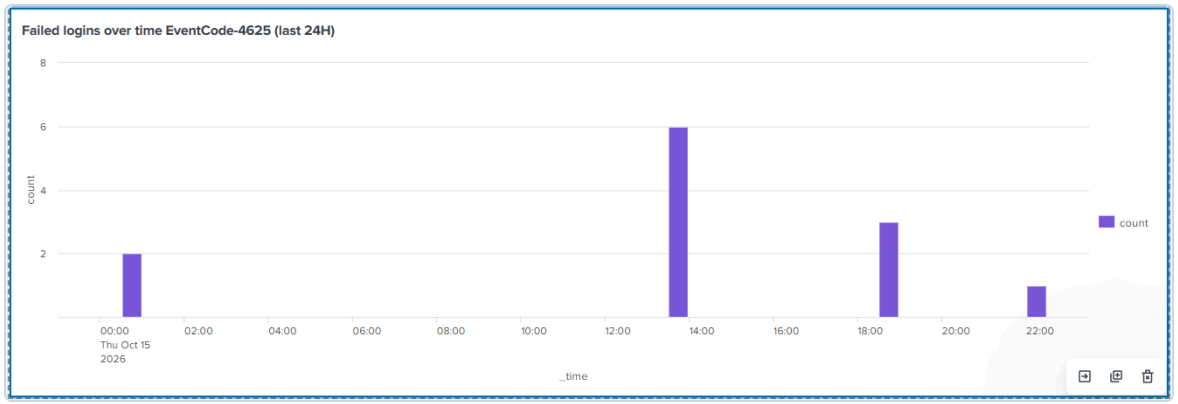


net user host

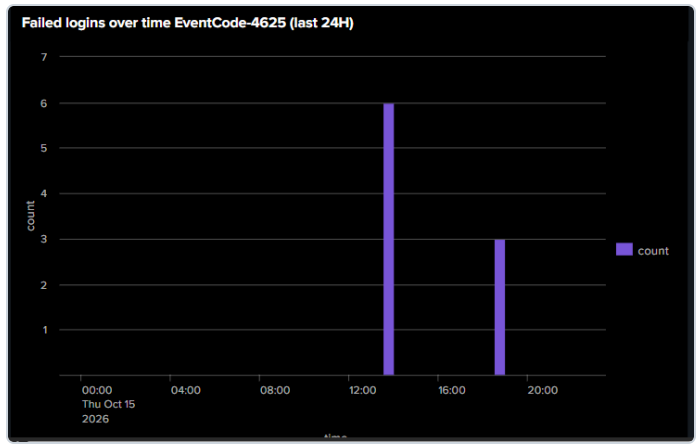


Get-LocalUser: SIDs, RID 1002 missing

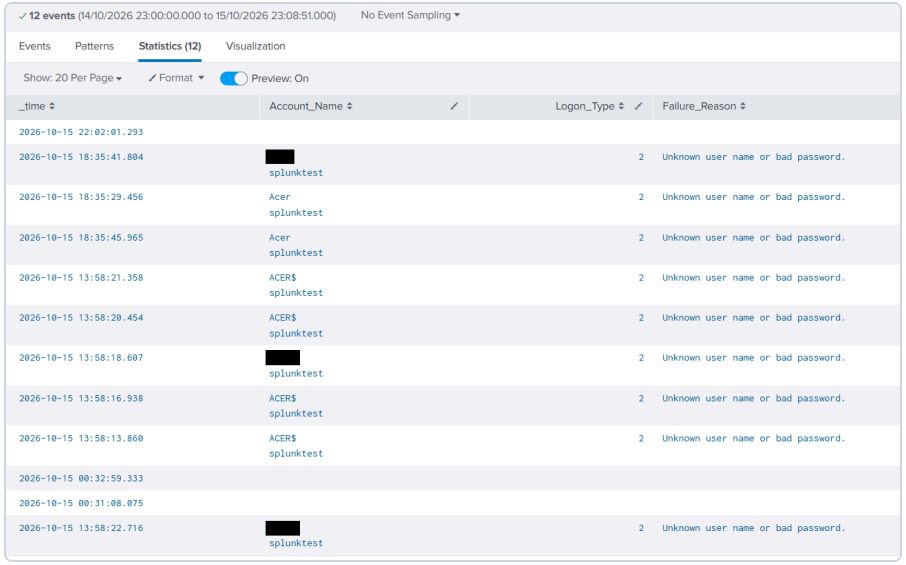
Lessons learned · False positives from my own search



Before: bars at 00:30 and 22:00 that weren't failed logins



After the fix: only the two real tests (13:58 and 18:35)



✓ **12 events** (14/10/2026 23:00:00.000 to 15/10/2026 23:08:51.000) No Event Sampling

Events | Patterns | **Statistics (12)** | Visualization

Show: 20 Per Page | Format | Preview: On

_time	Account_Name	Logon_Type	Failure_Reason
2026-10-15 22:02:01.293			
2026-10-15 18:35:41.804	[redacted] splunktest	2	Unknown user name or bad password.
2026-10-15 18:35:29.456	Acer splunktest	2	Unknown user name or bad password.
2026-10-15 18:35:45.965	Acer splunktest	2	Unknown user name or bad password.
2026-10-15 13:58:21.358	ACER$ splunktest	2	Unknown user name or bad password.
2026-10-15 13:58:20.454	ACER$ splunktest	2	Unknown user name or bad password.
2026-10-15 13:58:18.607	[redacted] splunktest	2	Unknown user name or bad password.
2026-10-15 13:58:16.938	ACER$ splunktest	2	Unknown user name or bad password.
2026-10-15 13:58:13.860	ACER$ splunktest	2	Unknown user name or bad password.
2026-10-15 00:32:59.333			
2026-10-15 00:31:08.075			
2026-10-15 13:58:22.716	[redacted] splunktest	2	Unknown user name or bad password.

Event 4625 results: 3 rows with empty fields

- My failed-login panel searched `EventCode=4625` across **every** log
- The blank rows were **Application**-log events from **EventSystem** using the same ID ('suppressing duplicate entries', logged at boot)
- **Event IDs aren't unique across logs**
- Fix: scope by log with `sourcetype="WinEventLog:Security"`

SECTION 4

Lessons learned

Splunk & searching

- **Field names are case-sensitive** (`EventCode`), but quoted text search isn't
- **Time range and span** decide whether a chart is empty, squashed or gapped
- **Collection ≠ display refresh:** the input's polling interval controls how often data is collected, while each dashboard panel has its own Search refresh interval. A normal search runs once; panels only update automatically if a refresh interval is set
- **Rare events matter most:** 4720, 4732 and 4740 happen too rarely to appear in a 'top 5' chart, so they need their own panel
- **A SIEM only knows what it collected:** Splunk held 2 days, but the host's own logs showed a problem going back to February

Windows logging

- **Sign-in throttling ≠ lockout policy:** 6 failed attempts triggered the sign-in screen's delay, but no 4740, because the policy only locks at its threshold (10 at the time)
- **One field, two accounts:** Windows events record both who did the action and who it was done to, and Splunk stores both in Account_Name. Counting by it without care can blame the wrong account
- **Event IDs aren't unique across logs:** a 4625 in the Application log (EventSystem) is a different event from a 4625 failed login in Security. Always scope by sourcetype
- **Low-privilege services need explicit rights:** Splunk's account needed Event Log Readers to read the Security log
- **Turning on 4688 multiplied the log volume,** mostly Splunk's own processes, so collection needs tuning
- **6008 times are approximate:** they show the last saved 'still alive' timestamp

Investigation method

- **Verify one sample across all events:** one TiWorker event suggested Windows Update, and `top Process_Name` proved it for all 7,789
- **Don't assume a setting from its events:** 4688s appeared in bursts at boot, but auditpol showed Process Creation was off. Only the configuration confirms a setting
- **Test one hypothesis at a time** and rule each out with evidence
- **Change one variable at a time:** updated only the graphics driver and left everything else unchanged, so any improvement can be attributed to the driver
- **Screenshot as you go, but check what's in them:** I left out screenshots showing device/licence IDs and a test password before sharing this write-up

SECTION 5

Recommendations & next steps

Host hardening

- Use a standard account day-to-day and a separate admin account only when needed
- Password policy: minimum length ≥ 12, keep password history, remove the never-expires exemption on host
- Keep the lockout threshold low (3–5)
- Disable or delete splunktest when the full project is complete

Stability (Finding 4)

- AMD driver updated (16 Oct); hibernate left unchanged so only one variable changed
- Monitor for new Event 41s over the next 2 weeks (ignore the 16 Oct 20:55 forced restart)
- If freezes continue: next single test is hibernate disabled (`powercfg /hibernate off`)

SIEM

- Alerts on 4740 (lockout) and 4732 into **Administrators**
- Brute-force alert: e.g. ≥ 5 × 4625 for one account within 5 minutes
- Monitor ingestion gaps: no events from a source = red flag
- Tune 4688: filter out Splunk's own processes to cut noise
- Add drilldowns so chart clicks open the underlying events